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Versatile, high sensitivity, and automatized angular dependent vectorial Kerr magnetometer for the analysis of nanostructured materials

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Magneto-optical Kerr effect (MOKE) magnetometry is an indispensable, reliable, and one of the most widely used techniques for the characterization of nanostructured magnetic materials. Information, such as the magnitude of coercive fields or anisotropy strengths, can be readily obtained from MOKE measurements. We present a description of our state-of-the-art vectorial MOKE magnetometer, being an extremely versatile, accurate, and sensitivity unit with a low cost and comparatively simple setup. The unit includes focusing lenses and an automatized stepper motor stage for angular dependent measurements. The performance of the magnetometer is demonstrated by hysteresis loops of Co thin films displaying uniaxial anisotropy induced on growth, MnIr/CoFe structures exhibiting the so called exchange bias effect, spin valves, and microfabricated flux guides produced by optical lithography.
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I. INTRODUCTION

Magneto-optical Kerr effect (MOKE) magnetometry is an efficient, nondestructive, and noninvasive technique for the study and characterization of magnetization reversal in thin films, sandwiches, and patterned magnetic elements because of its high sensitivity down to nanometer thickness.[1–11] The knowledge of the magnetization behavior is of crucial importance for the successful engineering of, e.g., high density magnetic hard disk drives, magnetic devices and memories.[12–15] Information, such as the magnitude of the coercivity, magnetic anisotropy, and interlayer exchange coupling can be accurately determined from the analysis of hysteresis loops recorded with a MOKE setup. Nonstandard optical techniques based on MOKE, such as time-resolved MOKE or Brillouin light scattering spectroscopy, allow for sensing magnetization dynamics on time scales down to femtosecond.[16]

MOKE can be regarded as a change in the intensity and/or polarization of light upon reflection from a magnetized material. The nature of the Kerr effect depends on the orientation of the magnetization (**M**) with respect to the incident and sample planes (Fig. 1).[17,18] When the magnetization is parallel to the sample surface and in the plane of incidence (M_l), we have the longitudinal Kerr effect (LMOKE), which manifests itself as a rotation of the polarization plane plus a phase shift which changes the incident linear polarized light into elliptically polarized (upon reflection). The same effect occurs if **M** is perpendicular to the sample surface, M_p (polar Kerr effect; PMOKE). On the other hand, for the transverse orientation, where **M** is parallel to the sample surface but perpendicular to the plane of incidence (M_t), only a change in the intensity of the reflected beam occurs (transversal Kerr effect; TMOKE). This last effect only happens when the light is polarized in the plane of incidence.

Here we report on a versatile, fast, sensitive, and easy to assemble vectorial MOKE magnetometry unit. It allows the simultaneous measurement of the two in-plane **M** components and thus to obtain the complete in-plane magnetization vector. We will describe in detail the implementation of this MOKE system, which is automatized with a rotatory motor stage to perform angular dependent measurements, allowing the extraction of magnetic anisotropy directions in thin films. It also incorporates focusing lenses to study magnetic patterned samples, as well as an x-, y-, and z-translation stage for positioning and sensing the magnetization in different regions of the sample. We exemplify the possibilities of the developed setup by probing Co thin films displaying uniaxial anisotropy induced on growth, MnIr/CoFe structures exhibiting the so-called exchange bias effect, spin valves, and microfabricated magnetic flux guides (FGs) produced by optical lithography.

II. MAGNETO-OPTICAL KERR EFFECT AND EXPERIMENTAL SETUP

A. In-plane vectorial MOKE

For a beam of light reflected from a magnetic material, one has that the amplitudes corresponding to polarization parallel (p) and perpendicular (s) to the plane of incidence are related to the incident p and s amplitudes by the Fresnel coefficients:[17]

$$\mathbf{E_r} = \mathbf{R} \cdot \mathbf{E_i}, \tag{1}$$

i.e.,

$$\begin{pmatrix} E_p \\ E_s \end{pmatrix}_r = \begin{pmatrix} r_{pp} & r_{ps} \\ r_{sp} & r_{ss} \end{pmatrix} \cdot \begin{pmatrix} E_p \\ E_s \end{pmatrix}_i, \tag{2}$$

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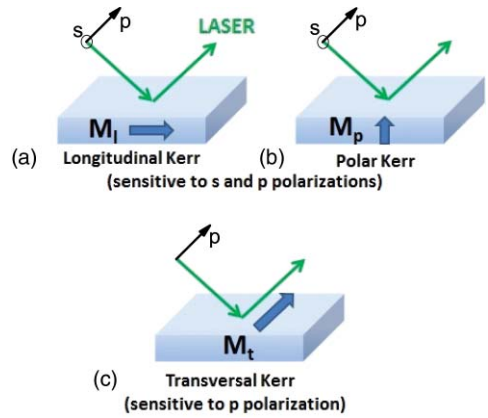

FIG. 1. (Color online) Kerr effect geometries. (a) Longitudinal, (b) polar, and (c) transversal. The longitudinal and polar geometries are both sensitive to the polarization of the incoming light parallel (p) and perpendicular (s) to the plane of incidence. The transversal geometry is sensitive only to the p-polarization.

where $\mathbf{E_r}$ is the reflected wave, $\mathbf{R}$ is a matrix with the Fresnel coefficients $r_{\alpha\beta}$ (with $\alpha, \beta = p, s$), and $\mathbf{E_i}$ is the incident wave. After performing some calculations (see Appendix), and for $\mathbf{E_i}$ polarized in the plane of incidence (E_p), we arrive to[17]

$$\mathbf{E_r} = E_i\left[\left(m_t^2 r_{pp}^t + m_l^2 r_{pp}^l + m_z^2 r_{pp}^p\right)\hat{\mathbf{p}} + \left(m_l^2 r_{sp}^l + m_z^2 r_{sp}^p\right)\hat{\mathbf{s}}\right], \tag{3}$$

where $\hat{p}$ and $\hat{s}$ are unit vectors defining the polarization directions parallel and perpendicular to the plane of incidence, respectively. The reflected intensity is therefore given by

$$I_r = |E_i|^2\left[\left|m_t^2 r_{pp}^t + m_l^2 r_{pp}^l + m_z^2 r_{pp}^p\right|^2 + \left|m_l^2 r_{sp}^l + m_z^2 r_{sp}^p\right|^2\right], \tag{4}$$

where $r_{\alpha\beta}^c$ are the reflection coefficients due to the polar ($c = p$), longitudinal ($c = l$), and transversal ($c = t$) Kerr effects for the electric field components parallel ($\alpha = p$ and/or $\beta = p$) and perpendicular ($\alpha = s$ and/or $\beta = s$) to the plane of incidence. The symbols m_t, m_l, and m_z refer to the relative magnitude (M/M_s; M_s is saturation magnetization) and direction of the magnetization components relative to the incident and sample planes.

In general, magneto-optical effects are small, being of fundamental interest to optimize the MOKE magnetometer to obtain maximum sensitivity. For the polar and longitudinal Kerr effects we achieve a good sensitivity on the Kerr rotation (θ_K) of the polarization plane by performing a differential measurement of the two components (E_s and E_p) of the reflected light. If Δ is the difference between the intensity of these two signals and $|\mathbf{E_r}|$ is the amplitude of the reflected light, we obtain

$$\Delta = |\mathbf{E_r}|^2(\cos^2\theta_K - \sin^2\theta_K) = |\mathbf{E_r}|^2\cos 2\theta_K, \tag{5}$$

and the sensitivity is therefore given by

$$\frac{d\Delta}{d\theta_K} = -2|\mathbf{E_r}|^2\sin 2\theta_K, \tag{6}$$

being maximum for $\theta_K = 45°$. Since the Kerr rotation angles are generally small ($<1°$), one should be able to, without changing the polarization state of the radiation, rotate its plane to $\approx 45°$. As will be seen below, this can be achieved using a half-wavelength ($\lambda/2$) wave plate.

The intensity of both p and s components emerging from the $\lambda/2$ wave plate will then be expressed by the equations

$$I_p = |\mathbf{E_r}|^2\cos^2(45° + \theta_K) = \tfrac{1}{2}|\mathbf{E_r}|^2(1 - \sin 2\theta_K), \tag{7}$$

and

$$I_s = |\mathbf{E_r}|^2\sin^2(45° + \theta_K) = \tfrac{1}{2}|\mathbf{E_r}|^2(1 + \sin 2\theta_K). \tag{8}$$

We immediately see that the Kerr rotation is proportional to the ratio between the difference ($I_s - I_p$) and the sum ($I_p + I_s$) of the two magneto-optical signals, according to

$$I_p + I_s = |\mathbf{E_r}|^2, \tag{9}$$

$$I_s - I_p = |\mathbf{E_r}|^2\sin 2\theta_K, \tag{10}$$

and therefore,

$$\theta_K = \frac{1}{2}\sin^{-1}\left(\frac{I_s - I_p}{I_p + I_s}\right). \tag{11}$$

Equation (10) highlights that the absence of Kerr rotation leads to $I_s = I_p$ and thus only the signal proportional to $I_p + I_s$ is different of zero. So, if there is no rotation in the plane of polarization of the beam but only a variation of its intensity, the sum of both signals will be sensitive to the transverse Kerr effect. For magnetic materials without an out-of-plane $\mathbf{M}$ component (as usual in thin films), $I_s - I_p$ arises only from the LMOKE, meaning that a measurement of both $I_s - I_p$ and $I_p + I_s$ signals allows the simultaneous detection of the two in-plane $\mathbf{M}$ components.

B. The implemented setup

Let us now consider the implemented experimental setup that allows us to take advantage of the above considerations (Fig. 2). In our setup a HeNe Laser produces a green (543.5 nm) light beam with a power of 4 mW. This beam passes through a Glan-Thompson Calcite polarizer with an extinction ratio of 10^{-5} to get linear polarized light parallel to the plane of incidence (E_p). Subsequently, the light hits the sample at $\sim 60°$ to its normal, to gain maximum reflection intensity due to the Fresnel equations. The sample is placed between the poles of an electromagnet, fixed on a bar which is mounted on a stepper motor stage, allowing the rotation of the sample and thus the automatization of angular dependent MOKE measurements. This enables to measure samples in a short time and without disturbing the measurement. The magnetic field (H) is applied in the plane of incidence and always parallel to the sample plane. The electromagnet is connected to an operational power supply (KEPCO BOP 100-4M) controlled by a signal generator (HP3245A), which enables to set the amplitude, frequency, and shape of H. To cool the electromagnet we use a small commercial fan. The stepper motor stage has an angular step precision of 5° and is fixed in a x-, y-, and z-translation stage (Thorlabs PT3/M; lateral resolution of 10 μm) for positioning the sample. It also allows sensing the magnetization in different regions of the sample. We can

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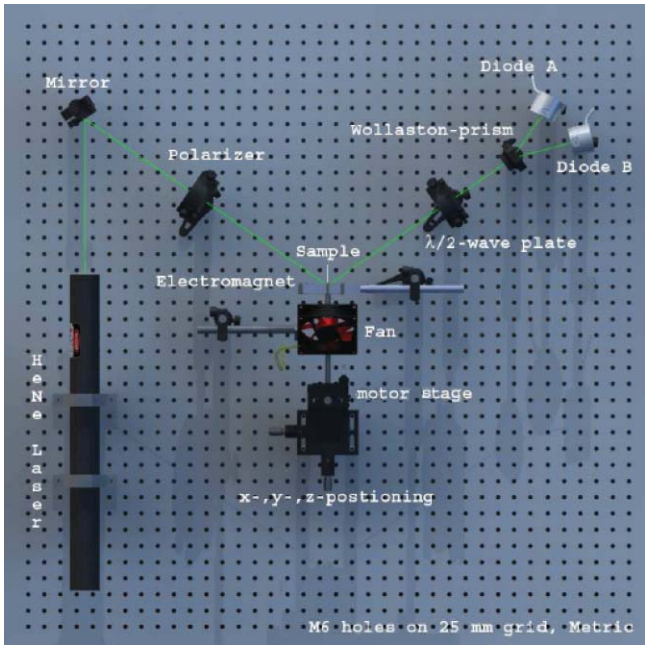

FIG. 2. (Color online) Scheme of the MOKE apparatus (Ref. 28). A detailed description of the devices is presented in the text.

also include a pair of optical lenses (Melles Griot, 5 cm of focus length; Fig. 3) to focus and collimate the laser beam before and after reflection, respectively. We are able to change the distance between the lenses and the sample and thus to vary the size of the laser spot down to 50μm. This possibility allows us to study magnetic patterned samples with μm dimensions.

Upon hitting the sample, the reflected light has a high electric field component in the plane of incidence (small θ_K) because the incident wave was initially linearly polarized in that plane (E_p). To maximize the sensitivity of the setup, we place (after the sample) a $\lambda/2$ wave plate with its optical axis rotated 22.5° relative to the p-plane. The beam will then emerge from the plate with its polarization plane rotated ~45° (2 × 22.5°) relative to the p-plane, since the reflected wave before crossing the $\lambda/2$ wave plate comes with its polarization plane rotated by θ_K, which is within minutes for most magnetic materials. Therefore, due to the Kerr rotation the polarization state fluctuates around the "average" direction defined at 45° of the p and s axes, depending on the direction and magnitude of the magnetization.

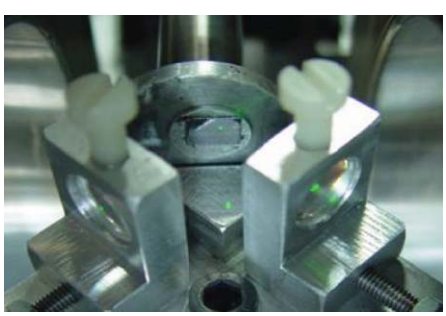

FIG. 3. (Color online) Photo of the optical pair lenses to focus and collimate the laser beam before and after reflection, respectively.

To measure the θ_K rotation, a Wollaston prism is used to separate the resulting beam into its two orthogonal components (E_p and E_s), which are then measured with two photodiodes (Fig. 2). A home made amplifier (up to 500×) was built to perform and amplify the difference and the sum of the two signals obtained from the two photodiodes. The Kerr and magnetic field signals are then displayed on an oscilloscope (Tektronix TDS 2024), which is connected to a computer via a GPIB interface.

The automatization of the MOKE magnetometer is performed through a home-made LABVIEW program (Fig. 4) which acquires and analyzes the data. It is organized in two tabs, the first dealing with stepper motor settings and the second with measurement settings. In the first tab, we can define, e.g., the initial orientation of the sample. In the second, the required settings include the angular steps and range, waveform (sinusoidal, triangular, square), H-amplitude, and frequency. The program also calculates the exchange (H_E) and coercive (H_c) fields of the measured samples. It has two displays, one shows the currently measured hysteretic cycle (left display; Fig. 4) and the other displays the angular dependence of H_E or H_c (right display). This has the advantage of an instantaneous overview of the measuring process.

III. MEASUREMENTS

The performance of the developed MOKE setup was tested on different magnetic samples: (A) continuous Co thin film, having an in-plane easy axis; (B) MnIr/CoFe bilayer sample exhibiting the exchange bias effect; (C) giant magnetoresistive spin valves; and (D) micropatterned CoZrNb flux guides. The measurements were made at room temperature in the longitudinal Kerr geometry with an applied field frequency of 1.4 Hz and a triangular waveform that provides a constant magnetic field sweep rate. To increase the signal-to-noise ratio, we average over 128 hysteresis loops. This is performed by the oscilloscope, so that one averaged cycle takes ~5 min to be acquired.

A. Co thin film

The Co thin films were prepared by an ion beam deposition (IBD) system from Commonwealth Scientific Corporation. The thin films were deposited on glass under a 5.0 sccm flow of Ar. The IBD chamber reaches a base pressure of less than 10^{-7} Torr and the working pressure during deposition was 2×10^{-4} Torr. The structure of the sample was Glass/Ta(5 nm)/Co(50 nm)/Ta(5 nm). During deposition, an in-plane magnetic field of 20 Oe was applied to define an uniaxial anisotropy in the samples. A detailed description of the growth process and deposition system can be found elsewhere.[19]

Figure 5 shows the $M(H)$ hysteretic cycles for the Co thin film in the longitudinal Kerr geometry. The square $M(H)$ loop with a sharp jump and the absence of hysteresis at 90° from that direction is indicative of an in-plane easy magnetization axis [Fig. 5(a)]. The measurements were performed by rotating the sample by an angle θ with respect to the plane of

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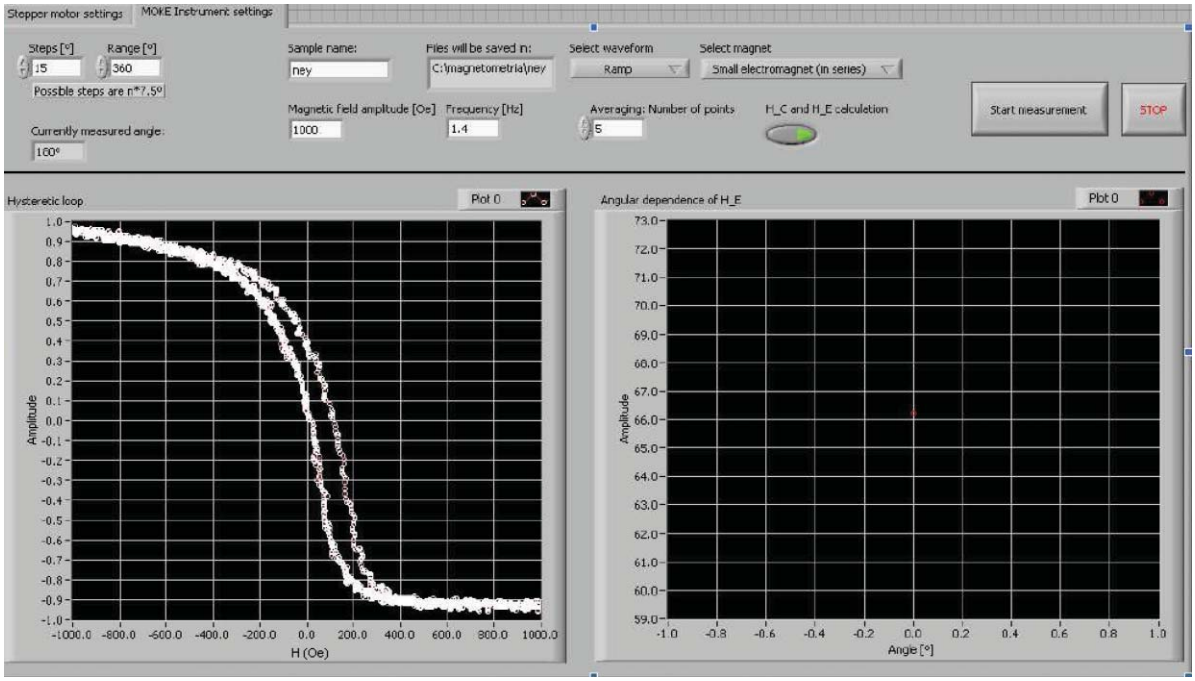

FIG. 4. (Color online) LABVIEW program for the automatization of the MOKE magnetometry unit.

incidence versus the H direction (always applied in the sample plane and parallel to the plane of incidence). The rotation was performed with the automatized motor stage. The angular dependence of H_c $(0-360°)$ is illustrated in the inset polar graph of Fig. 5(a). Here, the easy axis was made to coincide with $\theta = 0°$. The results were obtained from $\theta = 0°$–$180°$ and the $0-360°$ angular space was generated by mirroring the measured H_c values.[20] It is also shown that the two in-plane M components (M_l and M_t) are obtained simultaneously by the LMOKE and TMOKE at $\theta = 85°$ [Fig. 5(b)].

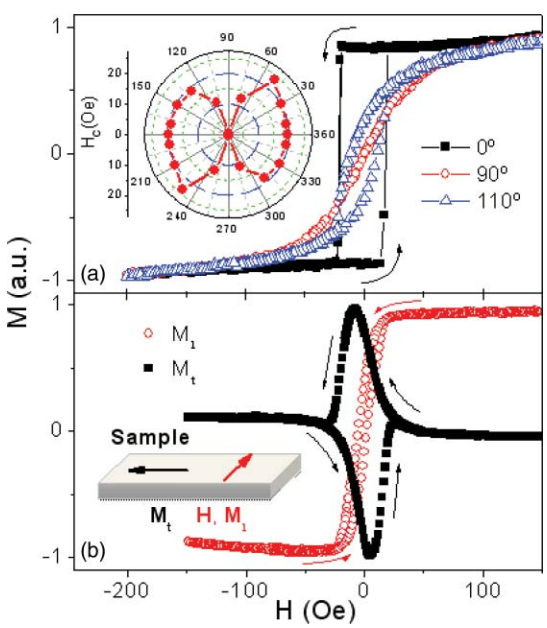

FIG. 5. (Color online) $M(H)$ hysteretic cycles of Co thin films obtained by the MOKE magnetometer. (a) Longitudinal component (M_L) for different orientations [$\theta = 0°$ (easy axis), 90° (hard axis), and 110°] of the sample. The inset polar graph shows the angular variation of H_c. (b) The two in-plane M (M_l and M_t) components obtained by the LMOKE and TMOKE at $\theta = 85°$.

B. MnIr/CoFe bilayer

We prepared the magnetic $Mn_{82}Ir_{18}$(4 nm)/$Co_{80}Fe_{20}$ (10 nm) bilayer samples with a NORDIKO 3000 IBD system. The base and deposition pressures of the system were $\sim 5 \times 10^{-8}$ and 2.4×10^{-4} mTorr, respectively. The deposition was performed using a Xe sputter gas with a flow rate of 4 sccm. To establish exchange bias the samples were annealed in vacuum ($<1 \times 10^{-5}$ mbar) at 493 K for 2 h, under a magnetic field of 3 kOe. The samples were grown on a glass substrate, covered (in the same run) by two successive 3 nm layers of Ru and Ta to ensure the subsequent growing of the antiferromagnetic MnIr phase. The CoFe layer deposition then followed, and finally a 10 nm Ta layer is added to prevent oxidation.

The H_E angular dependence of the MnIr(4 nm)/CoFe (10 nm) bilayer was extracted from the $M(H)$ loops obtained by our MOKE magnetometer. The corresponding cosinusoidal $H_E(\theta)$ dependence (Fig. 6) shows that the easy axis direction is at 0° and the hard direction is at 90°. For these angles H_E is maximal and minimal, respectively. The inset graph of Fig. 6 shows selected hysteretic cycles at $\theta = 0°$ and 30°.

C. Spin valves

On the same NORDIKO 3000 system, we also deposited spin valves[21] of composition Glass/Ta(3 nm)/NiFe (3 nm)/CoFe(2.5 nm)/Cu(2.1 nm)/CoFe(2.5 nm)/MnIr (8 nm) /Ta(2 nm). Here, NiFe, CoFe, and MnIr stand for $Ni_{80}Fe_{20}$, $Co_{80}Fe_{20}$, and $Mn_{82}Ir_{18}$.[22]

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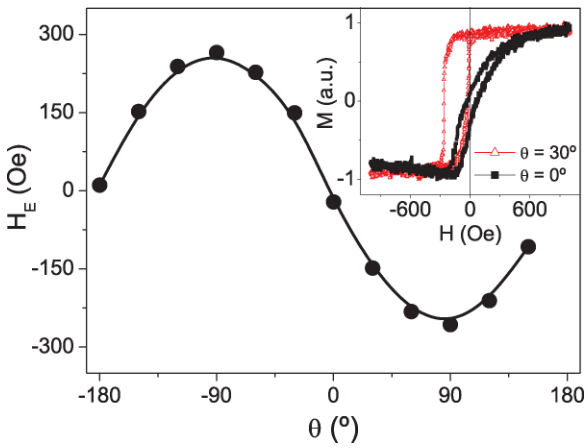

FIG. 6. (Color online) Angular dependence of H_E for the MnIr (4 nm)/CoFe(10 nm) bilayer. The inset graph shows selected MOKE hysteresis loops obtained at $\theta = 0°$ and 30°.

The high sensitivity of the MOKE system is demonstrated by comparing, for the same spin valve sample, the MOKE $M(H)$ loop (Fig. 7) with the one obtained by a superconducting quantum interference device (SQUID) magnetometer.[23] The low M values measured by SQUID ($\sim 10^{-5}$ emu) and the good signal-to-noise ratio of the MOKE $M(H)$ cycle illustrated in Fig. 7 clearly shows the very good sensitivity of our MOKE magnetometry unit. Assuming as a sensitivity criterium that the Kerr signal has to be at least as large as the noise level,[24] we estimate that the magnetic detection limit for our longitudinal MOKE setup is $\sim 10^{-7}$ emu.

D. Micropatterned CoZrNb flux guides

The capability of easily assembling a pair of optical lenses in the presented MOKE magnetometer to study micropatterned samples is here demonstrated for magnetic CoZrNb flux guides. A 500 nm thick $Co_{93.5}Zr_{2.8}Nb_{3.7}$ (at. %)

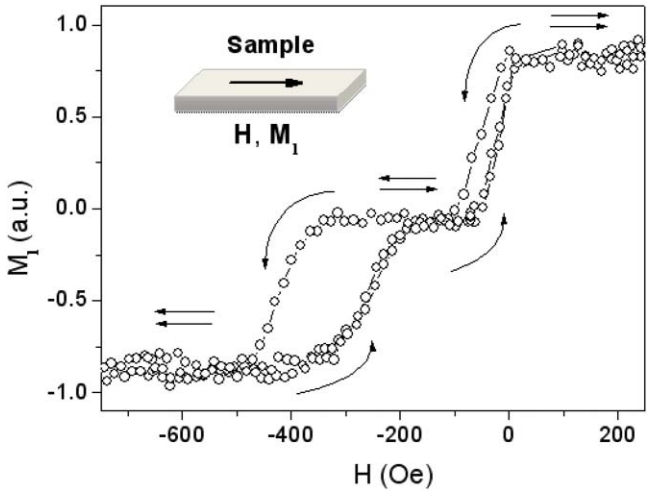

FIG. 7. (Color online) Easy axis $M(H)$ loop of a spin valve obtained by LMOKE at RT. The large signal-to-noise ratio enables a clear distinction of the two M reversal processes.

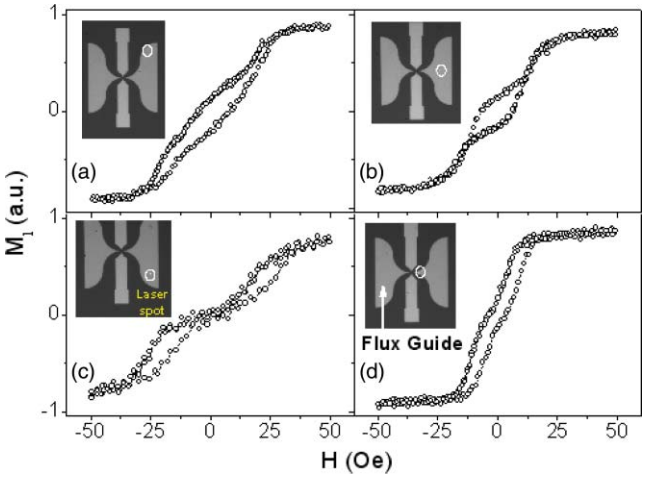

FIG. 8. (Color online) $M(H)$ Kerr effect loops, obtained with a focused laser beam with a spot diameter of $\sim 100\ \mu$m, incident in different regions of the CoZrNb FGs (inset photo). Laser spot on (a) top, (b) center, (c) down, and (d) near the spin valve sensor area of the FG.

soft thin film was micropatterned into a FG concentrator by a lift-off process, to convey and focus the external field into a spin valve sensor.[25–27] Figure 8 shows the patterned FG, having dimensions of $1000 \times 500\ \mu m^2$. The focused laser beam spot, with a diameter of $100\ \mu$m, was directed to the FG with the help of a CCD camera (VDS Vosskühler 1300) positioned in front of the sample plane. The x-, y-, and z-translation stage enabled moving the sample and thus to put the laser spot in the desired position. The MOKE hysteresis cycles were measured in different FG regions (Fig. 8), exhibiting changeable shapes with the laser spot position.

IV. CONCLUSIONS

We have illustrated a versatile, high sensitivity, and automatized angular dependent vectorial Kerr magnetometer, with a low cost and comparatively simple setup, for the study of nanostructured magnetic materials. The performance of the MOKE setup was demonstrated through the measurement of several magnetic samples. The possibility of easily assembling a pair of optical lenses to the system converts it in a focused beam MOKE magnetometer, allowing the analysis of micropatterned samples.

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APPENDIX: ADDITIONAL CALCULATIONS

In the following, we summarize the main steps leading to Eq. (3). According to Eqs. (1) and (2), the electrical field $\mathbf{E_r}$

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reflected by the magnetic sample can be calculated using the Fresnel reflection matrix **R**:

$$\mathbf{R} = m_t^2\mathbf{R^t}\left(\frac{Q}{m_t}\right) + m_l^2\mathbf{R^l}\left(\frac{Q}{m_l}\right) + m_z^2\mathbf{R^z}\left(\frac{Q}{m_z}\right) \quad \text{(A1)}$$

with $m_t = M_t/M$, $m_l = M_l/M$, $m_z = M_z/M$, and $\mathbf{R^t}$, $\mathbf{R^l}$, $\mathbf{R^z}$ being the Fresnel reflection matrices for the transversal, longitudinal, and polar Kerr effects. $\mathbf{E_r}$ is thus given by the expression

$$\begin{aligned}\mathbf{E_r} = {} & E_i\left[\left(m_t^2 r_{pp}^t + m_l^2 r_{pp}^l + m_z^2 r_{pp}^p\right)\cos\theta_p \right.\\ & \left. + \left(m_l^2 r_{ps}^l + m_z^2 r_{ps}^p\right)\sin\theta_p)\right]\hat{p}\\ & + E_i\left[\left(m_l^2 r_{sp}^l + m_z^2 r_{sp}^p\right)\cos\theta_p + r_{ss}^p\sin\theta_p\right]\hat{s}, \quad \text{(A2)}\end{aligned}$$

where θ_p is the angle of the polarizer transmission axis with the incident plane. Since $\mathbf{E_i}$ is polarized in the plane of incidence (E_p; $\theta_p = 0$), we arrive to Eq. (3) with the Fresnel coefficients given by

$$r_{pp}^t = \left(\frac{nA - B}{nA + B}\right)\left(1 + \frac{in^2 Q\sin 2\Theta}{n^2 m_t\left(n^2\cos^2\Theta - 1\right) + \sin^2\Theta}\right); \quad \text{(A3)}$$

$$r_{pp}^l = r_{pp}^p = \frac{nA - B}{nA + B}; \quad \text{(A4)}$$

$$r_{sp}^l = -\frac{in^2 QA\sin\Theta}{n^2 m_l B\,(nA + B)\,(A + nB)}; \quad \text{(A5)}$$

$$r_{sp}^p = -\frac{inQA}{m_z\,(nA + B)\,(A + nB)}, \quad \text{(A6)}$$

where $A = \cos\Theta$, $B = \sqrt{1 - \sin^2\Theta/n^2}$, Θ is the angle of incidence, Q is the Voigt magneto-optical parameter, and n is the refractive index. Note that the expressions presented for the Fresnel reflection coefficients are related to the air-film interface.

[1] J. Camarero, J. J. de Miguel, R. Miranda, and A. Hernando, J. Phys. Condens. Matter **12**, 7713 (2000).
[2] D. Atkinson, D. A. Allwood, M. D. Cooke, and R. P. Cowburn, J. Phys. D: Appl. Phys. **34**, 3019 (2001).
[3] D. A. Allwood, G. Xiong, M. D. Cooke, and R. P. Cowburn, J. Phys. D: Appl. Phys. **36**, 2175 (2003).
[4] J. Camarero, Y. Pennec, J. Vogel, M. Bonfim, S. Pizzini, F. Ernult, F. Fettar, F. Garcia, F. Lancon, L. Billard, B. Dieny, A. Tagliaferri, and N. B. Brookes, Phys. Rev. Lett. **91**, 027201 (2003).
[5] R. Morales, J. I. Martín, M. Vélez, and J. M. Alameda, Eur. Phys. J. B **40**, 463 (2004).
[6] V. Usov, S. Murphy, L. Seravalli, and I. V. Shvets, Rev. Sci. Instrum. **76**, 046102 (2005).
[7] W. Kleemann, Rev. Sci. Instrum. **78**, 120901 (2007).
[8] A. Westphalen, M.-S. Lee, A. Remhof, and H. Zabel, Rev. Sci. Instrum. **78**, 121301 (2007).
[9] D. A. Allwood, P. R. Seem, S. Basu, P. W. Fry, U. J. Gibson, and R. P. Cowburn, Appl. Phys. Lett. **92**, 072503 (2008).
[10] M. Buchmeier, R. Schreiber, D. E. Bürgler, and C. M. Schneider, Phys. Rev. B. **79**, 064402 (2009).
[11] E. Jimenez, J. Camarero, J. Sort, J. Nogues, N. Mikuszeit, J. M. Garcia-Martin, A. Hoffmann, B. Dieny, and R. Miranda, Phys. Rev. B. **80**, 014415 (2009).
[12] D. A. Allwood, G. Xiong, C. C. Faulkner, D. Atkinson, D. Petit, and R. P. Cowburn, Science **309**, 1688 (2005).
[13] O. Hellwig, A. Berger, T. Thomson, E. Dobisz, Z. Z. Bandic, H. Yang, D. S. Kercher, and E. E. Fullerton, Appl. Phys. Lett. **90**, 162516 (2007).
[14] J. M. Slaughter, Annu. Rev. Mater. Res. **39**, 277 (2009).
[15] W. R. Altman, J. Moreland, S. E. Russek, and V. M. Bright, J. Magn. Magn. Mater. **322**, 3236 (2010).
[16] D. Wang, A. Verevkin, R. Sobolewski, R. Adam, A. van der Hart, and R. Franchy, IEEE Trans. Nanotechnol. **4**, 460 (2005).
[17] J. M. Florczak and E. D. Dahlberg, J. Appl. Phys. **67**, 7520 (1990).
[18] J. M. Florczak, E. D. Dahlberg, J. N. Kuznia, A. M. Wowchak, and P. I. Cohen, J. Appl. Phys. **69**, 4997 (1991).
[19] R. Fermento, D. C. Leitao, J. M. Teixeira, A. M. Pereira, F. Carpinteiro, J. Ventura, J. P. Araujo, and J. B. Sousa, J. Non-Cryst. Solids **354**, 5279 (2008).
[20] R. Bali, N. A. Stelmashenko, and M. G. Blamire, J. Appl. Phys. **103**, 053911 (2008).
[21] A. Fert, Rev. Mod. Phys. **80**, 1517 (2008).
[22] J. M. Teixeira, J. O. Ventura, R. P. Fermento, J. P. Araujo, J. B. Sousa, S. C. Freitas, and P. J. Freitas, IEEE Trans. Magn. **43**, 3143 (2007).
[23] J. M. Teixeira, J. Ventura, R. Fermento, J. P. Araujo, J. B. Sousa, S. Cardoso, and P. P. Freitas, J. Appl. Phys. **103**, 07F319 (2008).
[24] A. Lehnert, P. Buluschek, N. Weiss, J. Giesecke, M. Treier, S. Rusponi, and H. Brune, Rev. Sci. Instrum. **80**, 023902 (2009).
[25] R. C. Chaves, P. P. Freitas, B. Ocker, and W. Maass, Appl. Phys. Lett. **91**, 102504 (2007).
[26] I. G. Trindade, J. Teixeira, R. Fermento, J. B. Sousa, S. Cardoso, R. C. Chaves, and P. P. Freitas, IEEE Trans. Magn. **44**, 2472 (2008).
[27] I. G. Trindade, D. C. Leito, Y. Pogorelov, J. B. Sousa, R. C. Chaves, S. Cardoso, and P. P. Freitas, Appl. Phys. Lett. **94**, 073501 (2009).
[28] See supplementary material at http://dx.doi.org/10.1063/1.3579497 for an animation of the developed experimental setup.

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